

Opto Diode Corporation Website

Hosting & Content Management Proposal

Prepared by: Engineering Team Date: April 14, 2026 Status: Ready for Review

Executive Summary

The new optodiode.com website has been built and is ready to go live. This document explains how the site will be hosted, how content will be updated day-to-day, and what the ongoing costs look like.

Key takeaways:

- The site can be live within a single business day
- Non-technical staff can add and edit products, blog posts, and resources without writing any code
- Content changes appear on the live site within seconds of clicking "Publish"
- Hosting and content management costs start at \$0/month on free tiers, scaling to ~\$35/month if traffic grows significantly
- No server maintenance, security patching, or IT overhead is required

What We Built

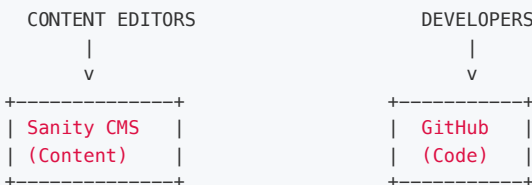
The website is a modern, high-performance application that includes:

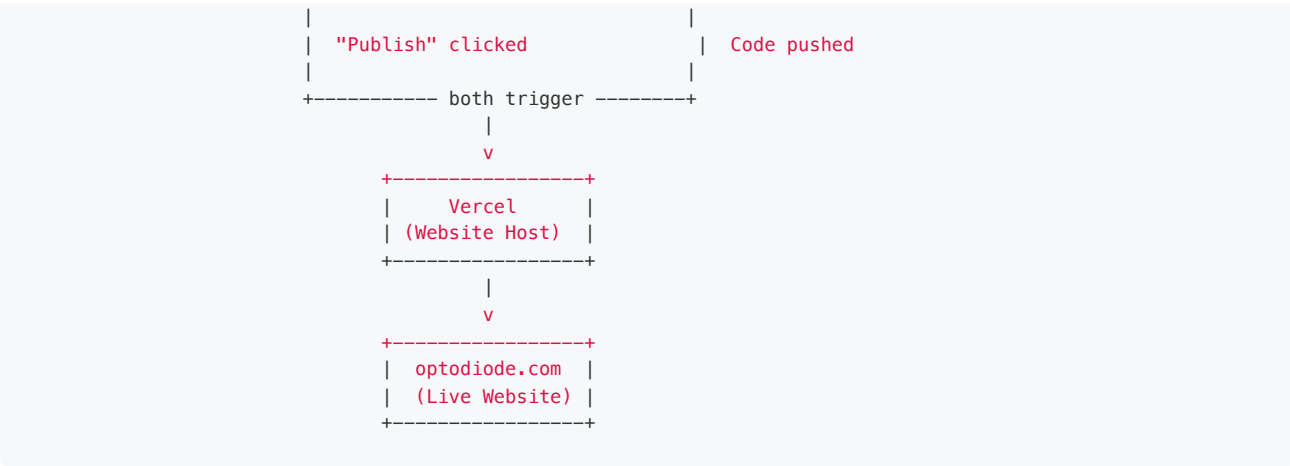
- **115 product pages** across Detection (photodiodes, IR detectors, APDs) and Emission (LEDs, IR emitters), each with full specifications, datasheet downloads, and related product links
- **Resource center** with blog, application notes, whitepapers, glossary, and compliance documentation
- **Industry-specific pages** for Semiconductor, Aerospace & Defense, Medical, and more
- **Built-in search** (Cmd+K command palette) across all products and resources
- **SEO optimization** with structured data for Google product listings
- **Mobile-responsive design** that works on all devices

How Hosting Works

The Recommended Approach

We recommend a three-service stack. Each service is an industry leader in its category, used by companies like Nike, Washington Post, Figma, and Sonos.





What Each Service Does

Vercel hosts the live website.

- Purpose-built for the technology our site uses (Next.js)
- Serves pages from data centers closest to each visitor (global CDN)
- Automatic HTTPS/SSL security certificates
- 99.99% uptime SLA on paid plans
- No servers to manage, patch, or monitor

Sanity is the content management system (CMS).

- Where staff log in to add products, edit descriptions, upload datasheets, and write blog posts
- Visual editing interface accessible from any web browser
- Changes publish to the live site in seconds, without waiting for a developer
- Full revision history — every change can be rolled back
- Role-based access control (editor, admin, viewer)

GitHub stores the website code.

- Industry-standard version control for the codebase
- Every change is tracked with full history
- Developers push code changes here; Vercel automatically deploys them

Day-to-Day Content Management

Who Does What

Task	Who	How	Time
Add a new product	Marketing / Engineering	Log into CMS, fill in form, click Publish	10-15 min
Edit product specs or descriptions	Marketing / Engineering	Find product in CMS, edit fields, click Publish	2-5 min
Upload a new datasheet PDF	Marketing / Engineering	Attach file in CMS product entry, click Publish	1-2 min

Task	Who	How	Time
Write a blog post	Marketing	Create new blog entry in CMS with rich text editor	30-60 min
Add an application note	Engineering	Create entry in CMS, attach PDF, click Publish	5-10 min
Change page layout or design	Developer	Edit code locally, push to GitHub, auto-deploys	Varies
Add a new product category	Developer	Code change + CMS schema update	1-2 hours

What Content Editing Looks Like

Staff access the CMS by visiting optodiode.com/studio in any web browser. No software installation is required. The interface presents forms tailored to each content type:

Product editing includes fields for:

- Product name and part number
- Description (short and long)
- Category and subcategory
- Specifications table (parameter / value / unit)
- Key specs (wavelength range, active area, responsivity, package)
- Features list
- Application areas
- Product image upload
- Datasheet PDF upload
- Related products (cross-linking)
- Featured / New product flags

Blog posts include:

- Title, author, publish date
- Rich text body with image embedding
- Tags and categories
- SEO metadata

All content types follow the same pattern: fill in the form, click Publish, and the change appears on the live site within seconds.

How Updates Reach the Live Site

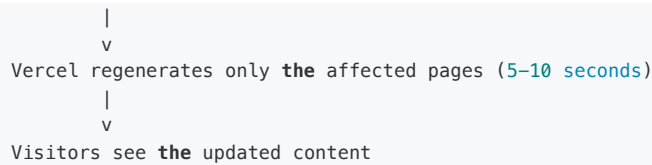
There are two paths for changes to reach the live website:

Path 1: Content Changes (No Developer Needed)

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Editor clicks "Publish" in CMS
      |
      v
Sanity sends a notification to Vercel

```



This process is called **Incremental Static Regeneration (ISR)**. Only the specific pages that changed are rebuilt — not the entire site. This means updates are fast and the site remains performant.

Path 2: Code Changes (Developer Required)



Every code push also creates a **preview URL** that can be reviewed before merging to production. This means design changes can be approved before going live.

Security & Reliability

Concern	How It's Handled
Site goes down	Vercel runs on AWS infrastructure across multiple regions. If one region fails, traffic routes to another automatically.
Someone publishes bad content	Sanity keeps full revision history. Any change can be rolled back to a previous version in one click.
Code breaks the site	Every deployment keeps the previous version. Instant rollback available in the Vercel dashboard.
Unauthorized access	CMS requires authenticated login. Role-based permissions control who can publish vs. draft.
SSL / HTTPS	Automatic. Vercel provisions and renews SSL certificates for all domains.
DDoS protection	Included with Vercel's edge network. No additional configuration needed.
Data backups	Sanity provides automated daily backups. Code is stored in GitHub with full history.

Cost Breakdown

Free Tier (Sufficient for Launch)

Service	Monthly Cost	What You Get
Vercel (Hobby)	\$0	100 GB bandwidth, unlimited deployments, SSL

Service	Monthly Cost	What You Get
Sanity (Free)	\$0	500K API requests, 20 GB bandwidth, 10 GB asset storage
GitHub (Free)	\$0	Unlimited private repositories
Total	\$0/month	

The free tiers are generous and will comfortably support the site at typical B2B traffic levels (thousands of visitors per month).

Growth Tier (If Traffic Scales)

Service	Monthly Cost	What You Get
Vercel (Pro)	\$20	1 TB bandwidth, team collaboration, analytics
Sanity (Growth)	\$15	2.5M API requests, higher bandwidth, advanced roles
GitHub (Free)	\$0	No change needed
Total	\$35/month	

Cost Comparison

For context, traditional hosting approaches for a site of this complexity:

Approach	Typical Monthly Cost	Maintenance
Recommended (Vercel + Sanity)	\$0 – \$35	None
Traditional shared hosting + WordPress	\$30 – \$100	Plugin updates, security patches, backups
AWS/Azure self-managed	\$100 – \$500	Server administration, monitoring, scaling
Managed enterprise CMS (Contentful, Adobe)	\$300 – \$3,000+	Minimal, but expensive

Timeline to Go Live

Step	Time Required	Who
Create accounts (GitHub, Vercel, Sanity)	15 minutes	Developer
Connect services and deploy	15 minutes	Developer
Point domain to new hosting	10 minutes + DNS propagation (up to 24h)	Developer + IT
Verify all pages and functionality	30 minutes	Developer + Marketing
Total active work	~1 hour	

The site can be live on a staging URL (e.g., `odc-website.vercel.app`) within the first 30 minutes. Pointing the production domain (`optodiode.com`) requires a DNS change that may take up to 24 hours to propagate globally, though it often completes within 1-2 hours.

What Happens to the Current Website

The transition can be handled with zero downtime:

1. Deploy the new site to a staging URL first
2. Review and verify everything works
3. When ready, update DNS to point `optodiode.com` to the new hosting
4. The old site stays available as a fallback until DNS fully propagates
5. Once confirmed, the old hosting can be decommissioned

Ongoing Maintenance

Task	Frequency	Effort
Content updates (products, blog, resources)	As needed	Staff self-service via CMS
Framework / dependency updates	Quarterly	1-2 hours (developer)
Security patches	Automatic	Vercel handles infrastructure security
SSL certificate renewal	Automatic	Vercel handles this
Backups	Automatic	Both Sanity and GitHub handle this
Server monitoring	Not applicable	No servers to monitor

Recommendation

We recommend proceeding with the Vercel + Sanity + GitHub stack because:

1. **It's ready.** The CMS integration, content schemas, and deployment pipeline are already built into the codebase. We only need to create accounts and connect them.
 2. **It empowers non-technical staff.** Products, blog posts, application notes, and other content can be managed through a visual editor without developer involvement.
 3. **It's cost-effective.** Free tiers cover launch and early growth. Even at scale, costs remain under \$35/month — a fraction of traditional hosting.
 4. **It eliminates maintenance burden.** No servers to manage, no security patches to apply, no backups to configure. The infrastructure is fully managed.
 5. **It's low-risk.** The transition involves zero downtime, every change is reversible, and the previous site remains available as a fallback.
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For questions or to authorize proceeding, contact the engineering team.